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Error estimates for the finite element method of the Navier-Stokes-Poisson-Nernst-Planck equations ⋆

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ABSTRACT

In this paper, we consider the linearized backward Euler finite element scheme for the Navier-Stokes-Poisson-Nernst-Planck (NSPNP) equations. We aim to derive the unconditional error estimates of all variables in $l^\infty(L^2)$ and $l^\infty(H^1)$ norms. Compared with some previous results, we remove the mesh restriction. In addition, we derive two important physical properties for the proposed numerical scheme. Finally, three numerical examples are implemented to verify the theoretical analysis.

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